Team :- 4 Review 1

Admin

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Project Title Hostel Mess Feedback & Voting System

A console-based Java application for student feedback and meal preference voting.

1. Project Description :

The Mess Feedback System is a Java-based console application designed to allow students to provide feedback for hostel mess meals and for admins to review all submitted feedback. The application uses basic File I/O operations to store and retrieve data from text files (users.txt and feedbacks.txt) and demonstrates principles of object-oriented programming, file handling, and menu-driven logic.

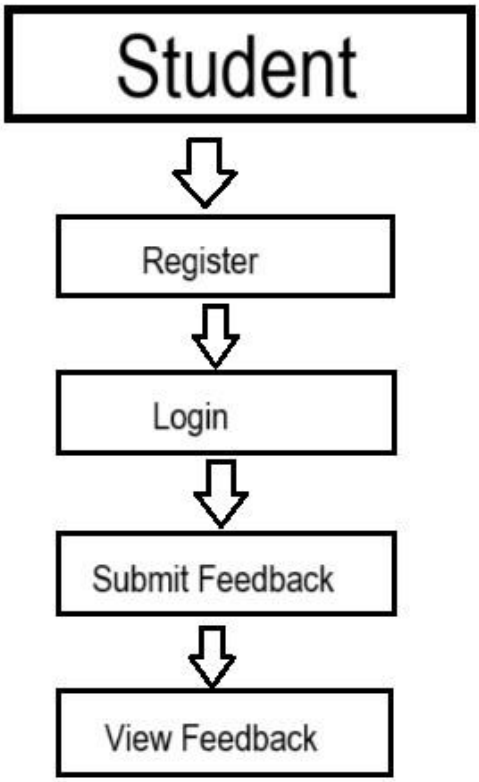
2. Objective

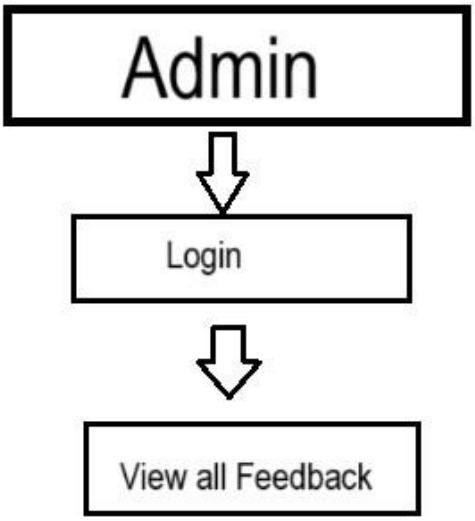
To build a simple and effective feedback management system for hostel mess using core Java without databases. The system will allow:
- User registration and login
- Role-based menu access (student/admin)
- Feedback submission with date and rating
- Viewing of submitted feedback (all or by user)

3. Technologies and Tools Used

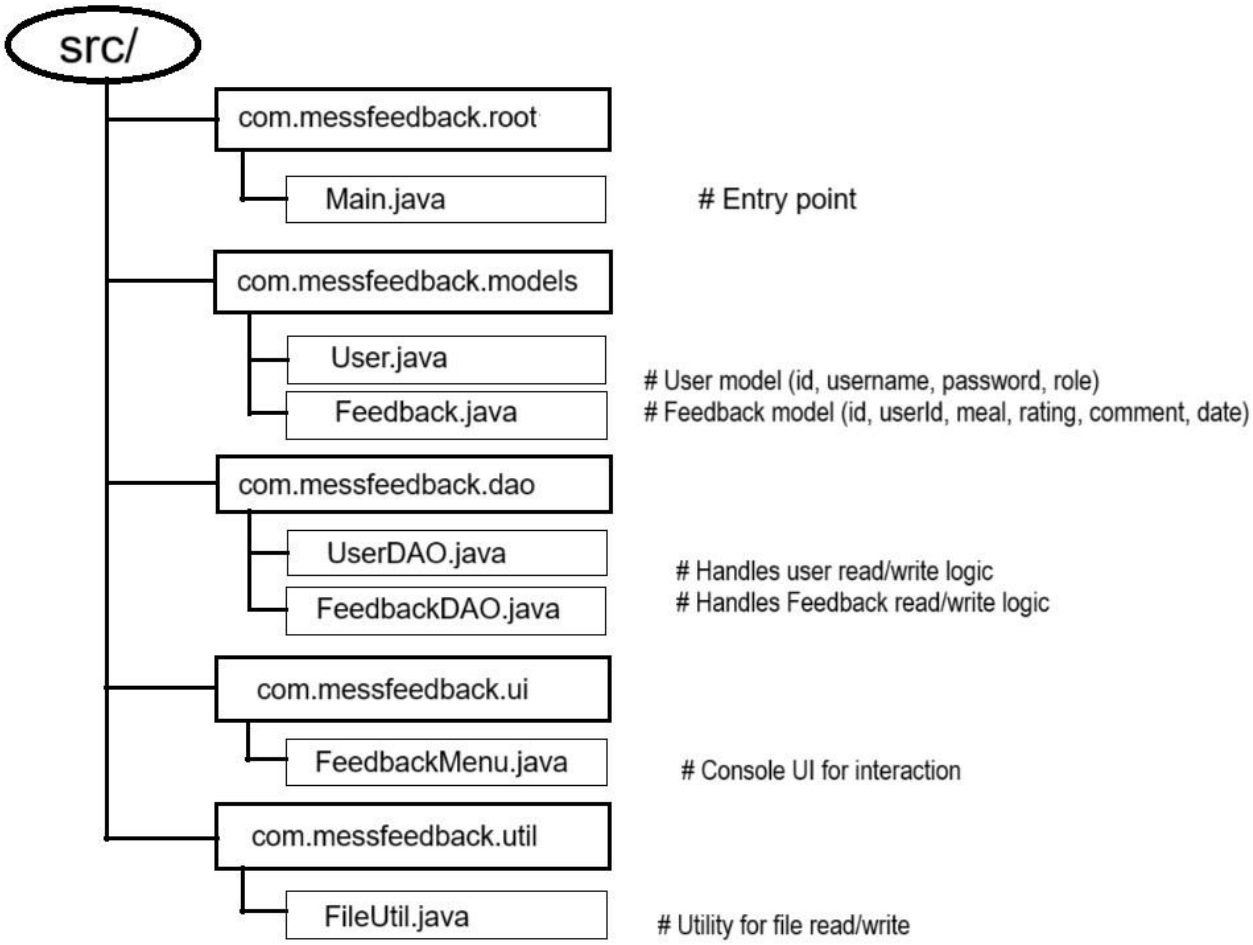
- Language: Java (JDK 17 or above)
- IDE/Text Editor: IntelliJ IDEA / Notepad
- File Handling: Text files (.txt) using Java I/O
- Architecture: Layered (Model, DAO, UI, Util)

4. Functional Requirements





5. Project Structure



1. models Package

Class: Feedback

Purpose:
This is the data model class used to represent feedback details submitted by a student.
Fields:
- int id – Unique identifier for each feedback.
- String studentName – Name of the student who gave feedback.
- String mealType – Type of meal (Breakfast/Lunch/Dinner).
- int rating – Rating given (typically from 1 to 5).

- String comments – Student's opinion or suggestion.
- String date – Date of feedback submission.

2. dao Package

Class: FeedbackDAO

Purpose:
This class handles all file-based CRUD operations for the feedback system.
Main Methods:
- addFeedback(Feedback fb) – Appends a new feedback to the file.
- List<Feedback> getAllFeedbacks() – Reads all feedback records from the file.
- boolean updateFeedback(int id, Feedback updatedFb) – Finds and updates feedback by ID.
- boolean deleteFeedback(int id) – Deletes a specific feedback entry by ID.

Note:
Uses FileWriter, BufferedReader, FileReader, etc. for file handling.

3. util Package (Optional but Useful)

Class: FileUtil

Purpose:
Provides reusable helper functions for file operations and data conversion.
Key Methods:
- int generateId() – Scans the file to return the next available feedback ID.
- String toCSV(Feedback fb) – Converts a Feedback object into a CSV line.
- Feedback fromCSV(String line) – Converts a CSV line back into a Feedback object.

4. ui Package

Class: FeedbackMenu

Purpose:
Contains all menu-related logic. Displays console-based menus and takes user input using Scanner.

Main Method:
- showMainMenu() – Displays options like Register, Login, Submit/View Feedback, etc., and handles user interaction.
Features:
- Displays user-friendly menu.
- Takes input from the user.
- Calls methods from FeedbackDAO to perform actions.
- Validates user inputs.

5. Root Package

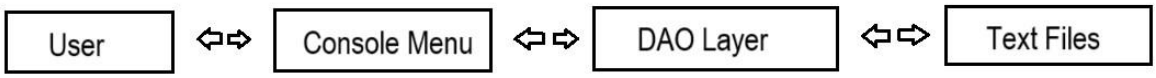
Class: Main

Purpose:
Acts as the entry point of the application.
Main Method:
- main(String[] args) – Instantiates FeedbackMenu and calls showMainMenu() to start the system.

Summary of Class Structure

Package	Class Name	Purpose
models	Feedback	Holds feedback fields (data model class)
dao	FeedbackDAO	Handles reading/writing to file (CRUD)
util	FileUtil	CSV conversion & ID generation (helper methods)
ui	FeedbackMenu	Console menu, user input, calls DAO methods
(root)	Main	Starts the application (calls menu method)

6. Data Flow Diagram



7. Features to Include

- **Create New Feedback (Write to file)**
- **View All Feedbacks (Read from file)**
- **Update Feedback (Edit file content)**
- **Delete Feedback (Remove entry from file)**
- **Input Validation**
- **Clear Console UI Design (Scanner-based)**

8. Sample File Structure

users.txt:
1,Bipin Nayak, Bipin@123, student
2,Deepak Singh,Deepak@123,admin

feedbacks.txt:
Feedback{feedbackId=1, userId=0, mealType='Breakfast', rating=1, comments='Its good in taste', date='2025-05-24'}

9. Screenshots

Registration

```
==== Mess Feedback System ====
1. Register
2. Login
3. Exit
Select an option: 1
Enter username: Bipin Nayak
Enter password: Bipin@123
Enter role (student/admin): student
Registration failed. Username might already exist.
```

Login

```
==== Mess Feedback System ====
1. Register
2. Login
3. Exit
Select an option: 2
Enter username: Bipin Nayak
Enter password: Bipin@123
Login successful!
```

Feedback submission

```
Welcome, Bipin Nayak
1. Submit Feedback
2. View My Feedback
3. Logout
Choice: 1
Meal Type (Breakfast/Lunch/Dinner): Lunch
Rating (1-5): 4
Comments: Its little bit good not bad in taste
Feedback submitted successfully.
```

Student view there submission

```
Welcome, Bipin Nayak
1. Submit Feedback
2. View My Feedback
3. Logout
Choice: 2
Your Feedbacks:
Feedback{feedbackId=1, userId=0, mealType='Breakfast', rating=1, comments='Its good in taste', date='2025-05-24'}
Feedback{feedbackId=2, userId=0, mealType='Lunch', rating=4, comments='Its little bit good not bad in taste', date='2025-05-24'}
```

Student logout

```
Welcome, Bipin Nayak
1. Submit Feedback
2. View My Feedback
3. Logout
Choice: 3
```

Admin login

```
==== Mess Feedback System ====
1. Register
2. Login
3. Exit
Select an option: 2
Enter username: Deepak Singh
Enter password: Deepak@123
Login successful!
```

Admin View

```
Welcome Admin, Deepak Singh
1. View All Feedback
2. Logout
Choice: 1
All Feedbacks:
Feedback{feedbackId=1, userId=0, mealType='Breakfast', rating=1, comments='Its good in taste', date='2025-05-24'}
Feedback{feedbackId=2, userId=0, mealType='Lunch', rating=4, comments='Its little bit good not bad in taste', date='2025-05-24'}
```

Admin Logout

```
Welcome Admin, Deepak Singh
1. View All Feedback
2. Logout
Choice: 2
```

10.Known Issue: Input Mismatch Exception

During the program execution, if the user **enters an invalid input type** (e.g., typing a word like "register" instead of the number 1), the application throws the following exception:

```
==== Mess Feedback System ====
1. Register
2. Login
3. Exit
Select an option: register
Exception in thread "main" java.util.InputMismatchException Create breakpoint
    at java.base/java.util.Scanner.throwFor(Scanner.java:964)
    at java.base/java.util.Scanner.next(Scanner.java:1619)
    at java.base/java.util.Scanner.nextInt(Scanner.java:2284)
    at java.base/java.util.Scanner.nextInt(Scanner.java:2238)
    at com.messfeedback.ui.FeedbackMenu.start(FeedbackMenu.java:31)
    at com.messfeedback.Main.main(Main.java:11)
```

- **Reason:**

The application expects numeric input using Scanner.nextInt() for menu selections (like 1, 2, 3). If the user types text such as "register" instead of selecting a number, it causes a **type mismatch**.

- **How It Can Be Handled (Future Scope):**

- Add **input validation** using Scanner.hasNextInt() before reading an integer.
- Wrap nextInt() in a try-catch block to handle incorrect input and prompt the user again instead of crashing.

- **Temporary User Instruction:**

For stable operation, users should **enter numeric options (1, 2, 3, etc.)** exactly as shown in the menu.

11.Challenges Faced

- Handling input mismatch exceptions using Scanner.nextInt()

- Ensuring proper file format and parsing comments containing commas

- Assigning unique feedback IDs based on file contents

12. Conclusion

This project helped in learning:

- Java File I/O operations (read/write text files)

- Modular programming using models and DAOs

- Exception handling and user input validation

- Implementing real-world logic using simple console interaction